

Open Science in Practice: Collaborative Research with Git



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HT2026



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Mini-project and data overview

Niklas Edvall

Requirements for mini-project

- Write a pipeline to analyze a dataset
- Minimum analysis is one figure + one "result"
- Data goes in → Results (report) comes out
- In a public repo on GitHub
- Plausible commit history
- At least one (new) branch + merge
- Well organized – no monoliths
- Well documented (README) and commented

https://github.com/NiklasEdvall/an-approved-repo

 **an-approved-repo** Public

 Pin  Watch **0**  Fork **0**  Star **0**

 main  1 Branch  0 Tags

 Go to file  Add file  Code

 **NiklasEdvall** Revise README for clarity and organization 6532268 · 4 minutes ago 25 Commits

 data	simulated data and code key	4 hours ago
 scripts	added some descriptive text to report template	10 minutes ago
 .gitignore	gitignore	1 hour ago
 README.md	Revise README for clarity and organization	4 minutes ago
 run_pipeline.R	master script to run pipeline	1 hour ago

 README

an-approved-repo

This is an example of an organized repo for the course [Open Science in Practice: Collaborative Research with Git](#).

Purpose

- Create a simulated dataset of 150 patients with either no, mild or severe tinnitus, some basic demographics and results from a hearing test (pure tone audiometry + UCL).
- Clean up, visualize and summarize the simulated data in an organized and well described pipeline.
- Use quarto markdown template to generate a summary report as html

Dependencies

About

This is supposed to be a good example of a well organized repo for the course "Open Science in Practice: Collaborative Research with Git"

 Readme

 Activity

 0 stars

 0 watching

 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

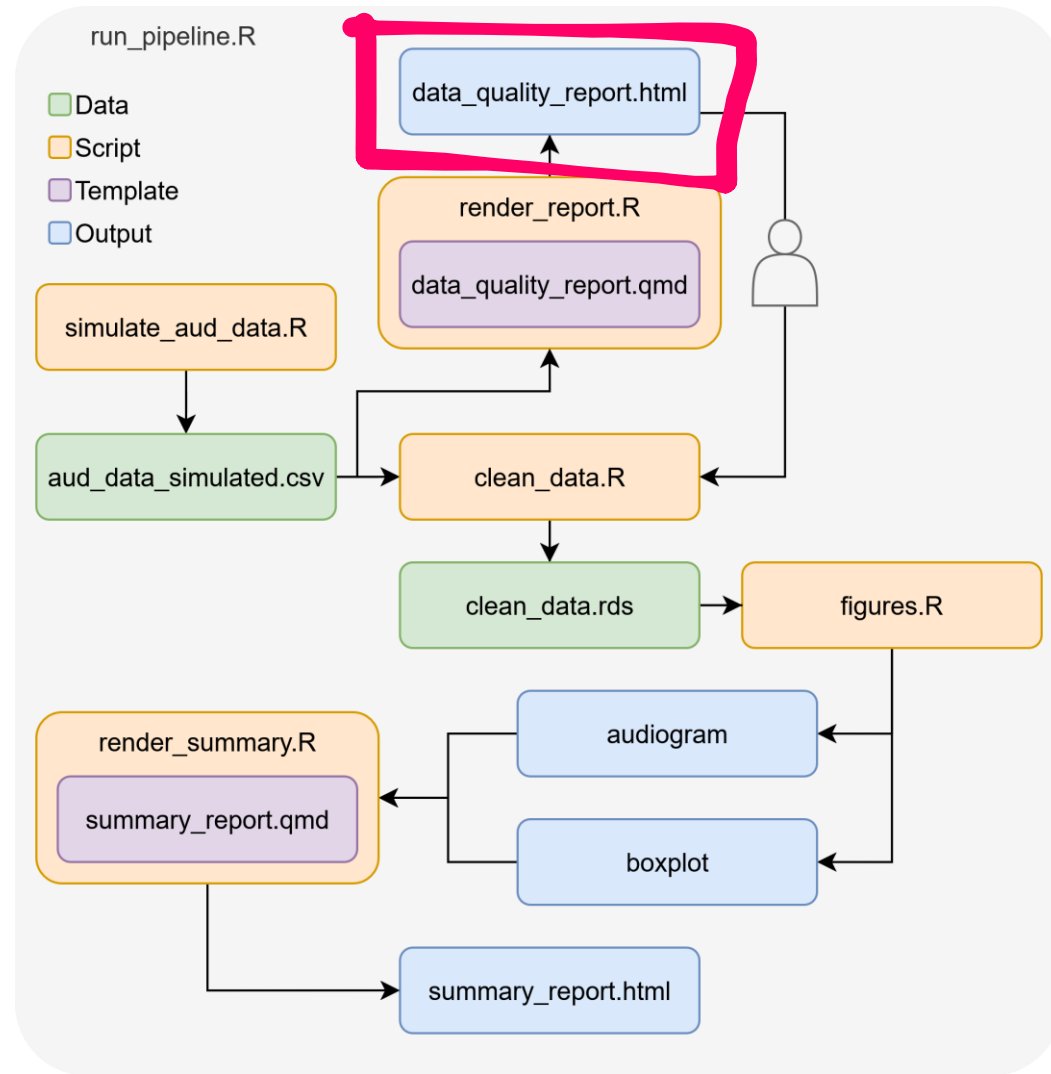
Contributors

1

 **NiklasEdvall**

Languages

 **R** 100.0%



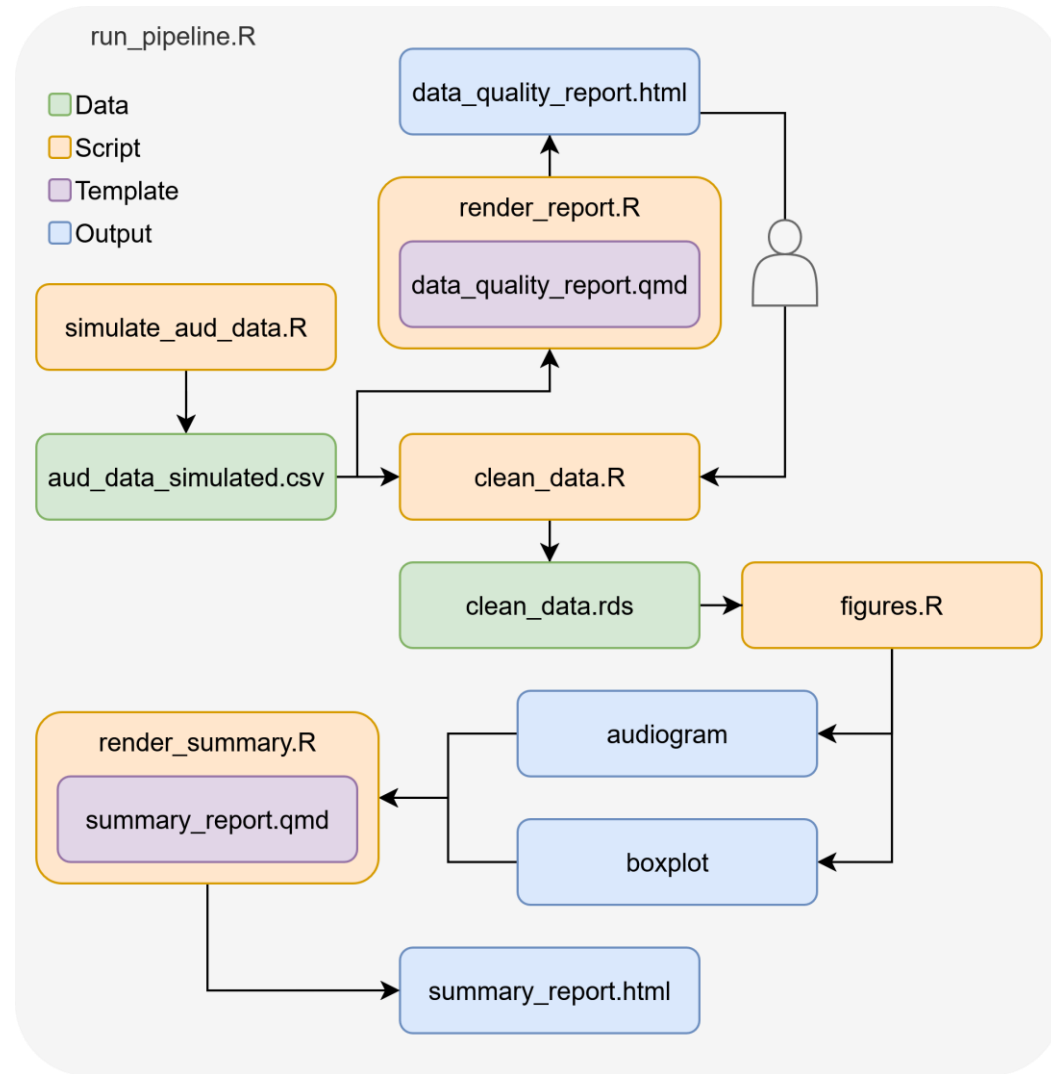
Audiological Data — Quality Report

PUBLISHED
September 2, 2026

Descriptive table by tinnitus group

Data stratified by tinnitus severity group in table to highlight any data quality issues.

	No tinnitus (N=59)	Mild tinnitus (N=52)	Severe tinnitus (N=39)	Total (N=150)
Age				
Median	39.0	39.5	41.0	40.0
Range	21.0 - 63.0	21.0 - 64.0	21.0 - 2000.0	21.0 - 2000.0
Sex				
Male	24 (40.7%)	24 (46.2%)	29 (74.4%)	77 (51.3%)
Female	34 (57.6%)	27 (51.9%)	10 (25.6%)	71 (47.3%)
Other	1 (1.7%)	1 (1.9%)	0 (0.0%)	2 (1.3%)
L125				
Median	5.0	9.0	8.0	7.0
Range	-8.0 - 19.0	-2.0 - 27.0	-7.0 - 21.0	-8.0 - 27.0
L250				
Median	6.0	9.5	9.0	8.0



Descriptive statistics by tinnitus group

Mean and standard deviation (SD) are reported for continuous variables, while counts and percentages are reported for categorical variables.

Two noteworthy observations:

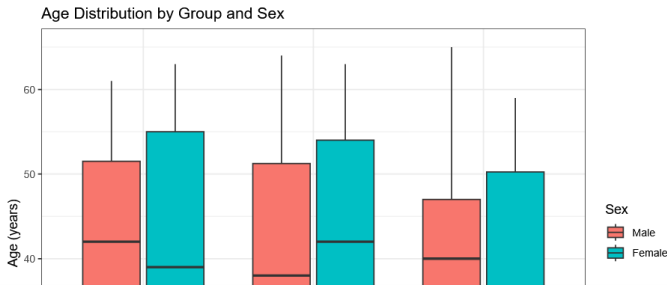
- Men are over-represented in the severe tinnitus group (74.4%)
- Hearing thresholds are around 10 dB higher in the severe tinnitus group compared to the group with no tinnitus.

Descriptive statistics by tinnitus severity group

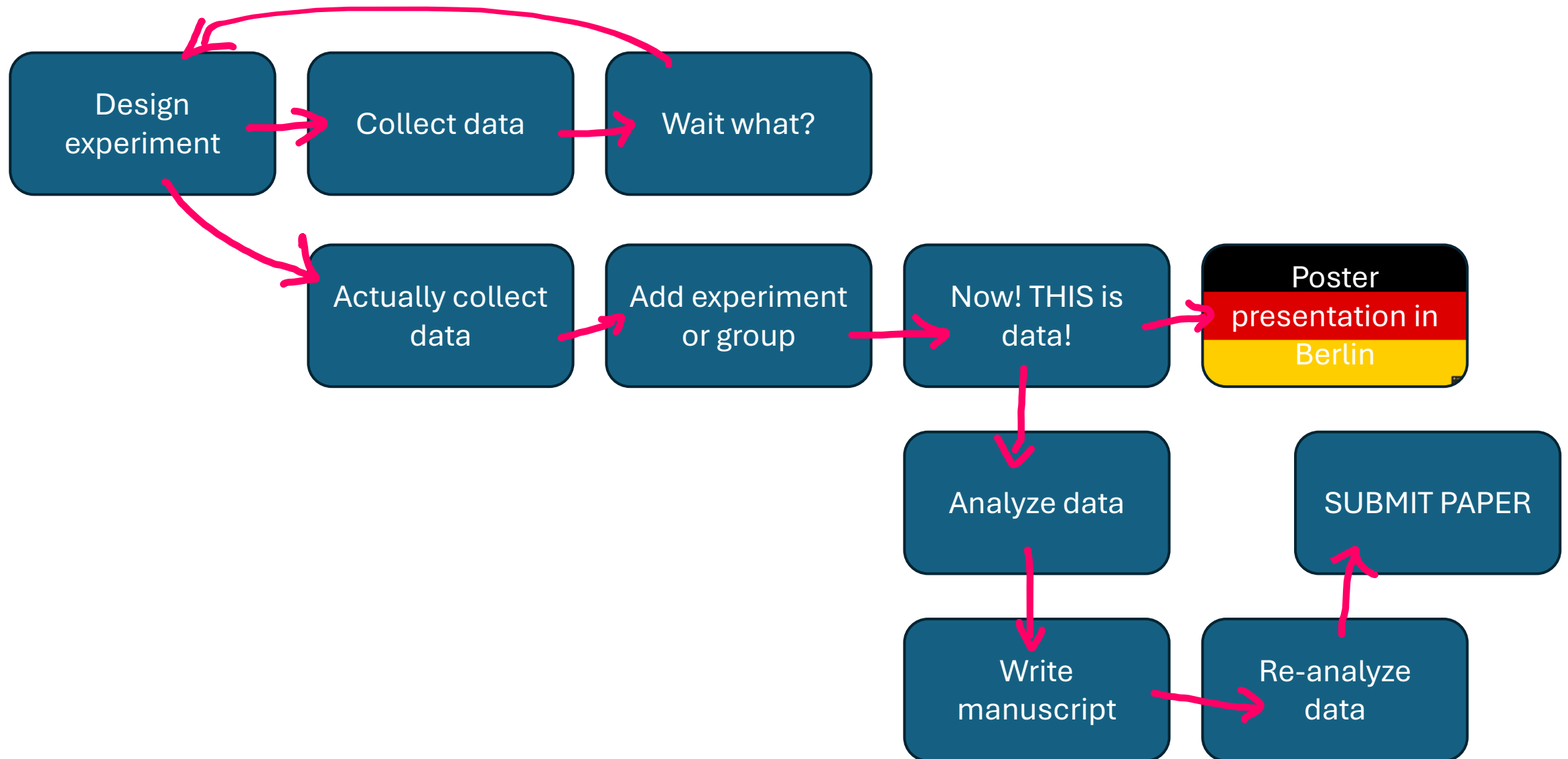
	No tinnitus (N=58)	Mild tinnitus (N=52)	Severe tinnitus (N=39)	Total (N=149)
Age				
Mean	40.1	41.3	39.4	40.4
SD	13.3	12.5	13.1	12.9
Range	21.0 - 63.0	21.0 - 64.0	21.0 - 65.0	21.0 - 65.0
Sex				
Male	24 (41.4%)	24 (46.2%)	29 (74.4%)	77 (51.7%)
Female	33 (56.9%)	27 (51.9%)	10 (25.6%)	70 (47.0%)
Other	1 (1.7%)	1 (1.9%)	0 (0.0%)	2 (1.3%)
PTA4L				
Mean	9.6	13.2	18.2	13.1
SD	3.4	3.3	3.7	4.9
Range	2.5 - 16.5	4.5 - 21.0	10.0 - 26.5	2.5 - 26.5
PTA4R				
Mean	10.3	13.3	17.8	13.3
SD	3.3	4.7	3.8	4.9
Range	3.2 - 19.0	5.2 - 25.5	10.8 - 24.5	3.2 - 25.5

Age distribution by group and sex

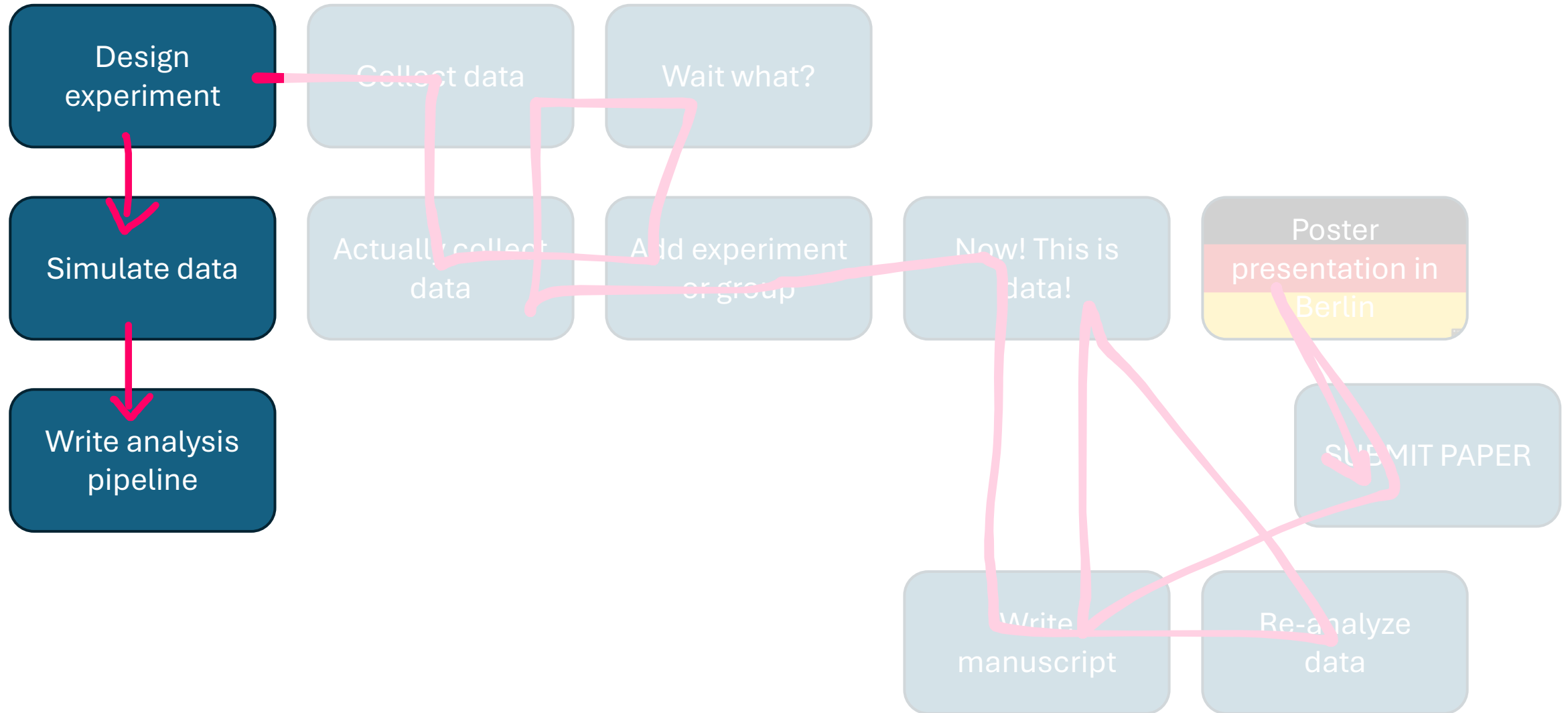
The boxplot below displays the age distribution by tinnitus severity group, for men and women separately. Sex reported as "Other" was only reported by four subjects and is excluded from this plot due to too few observations for meaningful interpretation. The median age is similar across groups.



Simulating data



Simulating data



Dataset 1

Survey of staff and students energy-levels and general well-being.

Items

- Sex, age, position, screen time, sleep,
- WHO-5: The World Health Organization-Five Well-Being Index
- PHQ-4: Patient Health Questionnaire for Depression and Anxiety

To-dos:

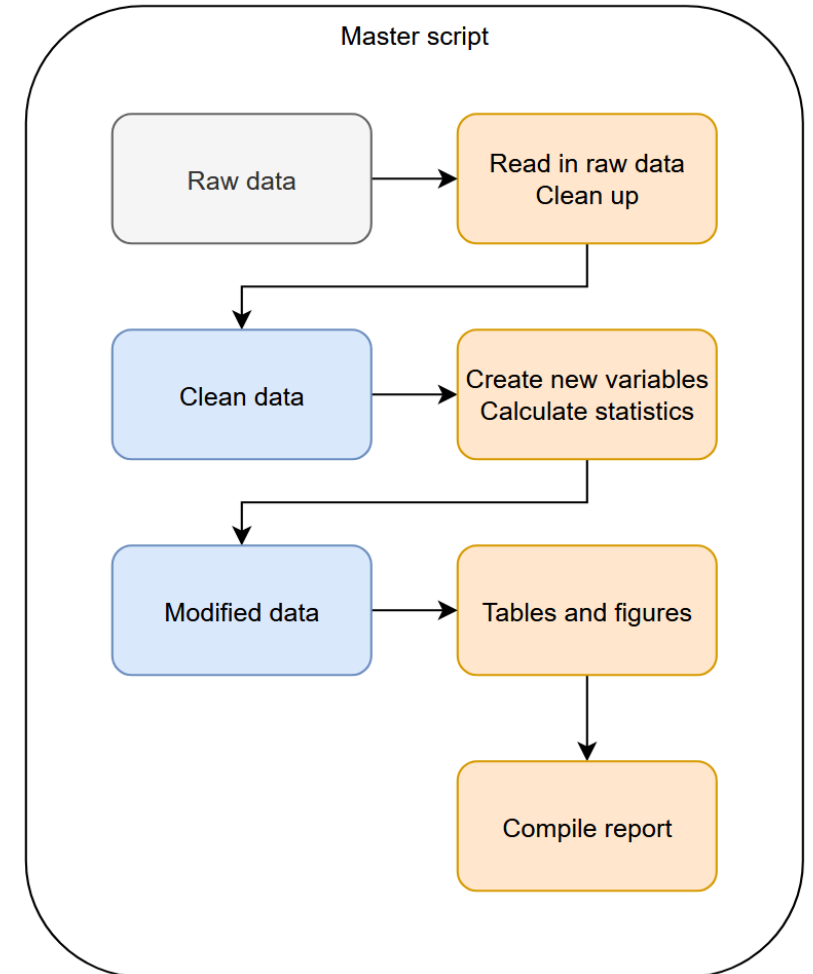
- Clean up data
- Calculate questionnaire scores
- Look for something to report

For example:

Are teachers or students happier?

Do men or women sleep better?

Is there a relationship between sleep and well-being?



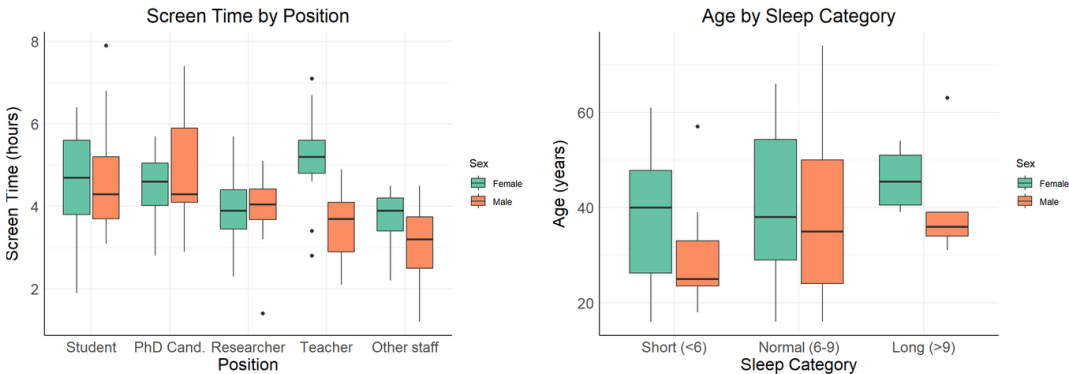
Survey recruitment methods showed differential effectiveness, with the majority of participants (59%) recruited through the newsletter, followed by lecture announcements (23%) and colleague referrals (16%). A small proportion of missing data (2.5%) was observed for the survey recruitment variable.

	Student (N=39)	PhD Cand. (N=15)	Researcher (N=26)	Teacher (N=25)	Other staff (N=15)	Total (N=120)
Age (years)						
Mean (SD)	24.3 (7.8)	34.5 (8.3)	52.8 (11.2)	42.9 (9.0)	46.2 (12.6)	38.4 (14.6)
Survey						
Webpage	2 (5.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (1.7%)
Newsletter	20 (52.6%)	12 (80.0%)	17 (65.4%)	12 (48.0%)	8 (61.5%)	69 (59.0%)
Colleagues	9 (23.7%)	1 (6.7%)	2 (7.7%)	5 (20.0%)	2 (15.4%)	19 (16.2%)
Lecture	7 (18.4%)	2 (13.3%)	7 (26.9%)	8 (32.0%)	3 (23.1%)	27 (23.1%)
N-Miss	1	0	0	0	2	3

Descriptives

Figure 1 illustrates the distribution of daily screen time across different academic positions, stratified by sex. Overall, screen time varied substantially within each position category, with students showing the highest median screen time (around 4.5 hours/day), followed by PhD candidates and researchers. Notable sex differences emerged across most positions, with males generally reporting higher screen time than females.

The second panel examines age in relation to sleep duration categories, revealing a relationship where individuals with shorter sleep duration (<6 hours) tended to be younger, while those with longer sleep (>9 hours) have celebrated more birthdays.



Results

Figure 2 presents the relationships between age and mental health outcomes. The left panel shows a modest positive association between age and WHO-5 well-being scores ($r = 0.185$), suggesting that older participants reported slightly higher subjective well-being.

Conversely, the right panel reveals a weak negative relationship between age and PHQ-4 scores ($r = -0.123$), indicating a trend toward lower anxiety and depression symptoms with increasing age, though this association did not reach conventional statistical significance. These patterns align with existing literature suggesting potential age-related differences in mental health outcomes within academic settings.



Dataset 2

EEG during n-back task pre and post training.

Items:

dataset-2-key.xlsx

group_data.csv – ID, sex, age, training days completed

sub-01_ses-01_task-nback_behav.csv

sub-01_ses-01_task-nback_eeg.csv



Dataset 2

EEG during n-back task pre and post training.

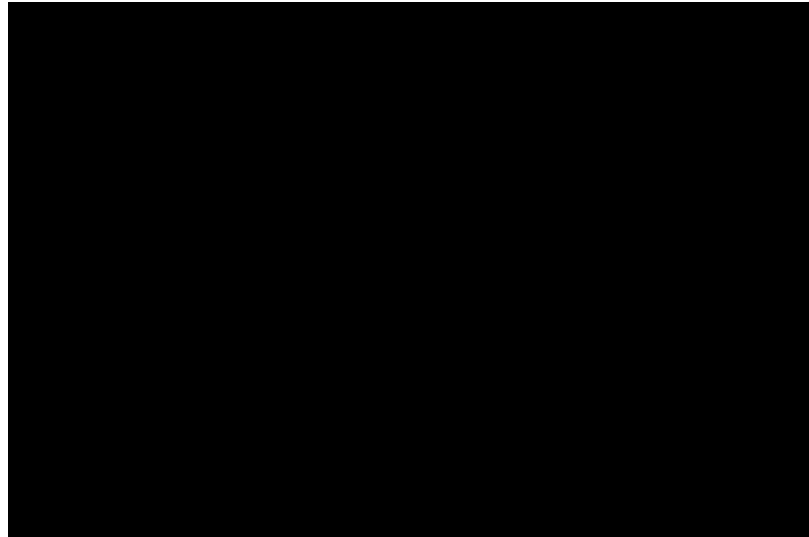
Items:

dataset-2-key.xlsx

group_data.csv – ID, sex, age, training days completed

sub-01_ses-01_task-nback_behav.csv

sub-01_ses-01_task-nback_eeg.csv



Dataset 2

EEG during n-back task pre and post training.

Items:

dataset-2-key.xlsx

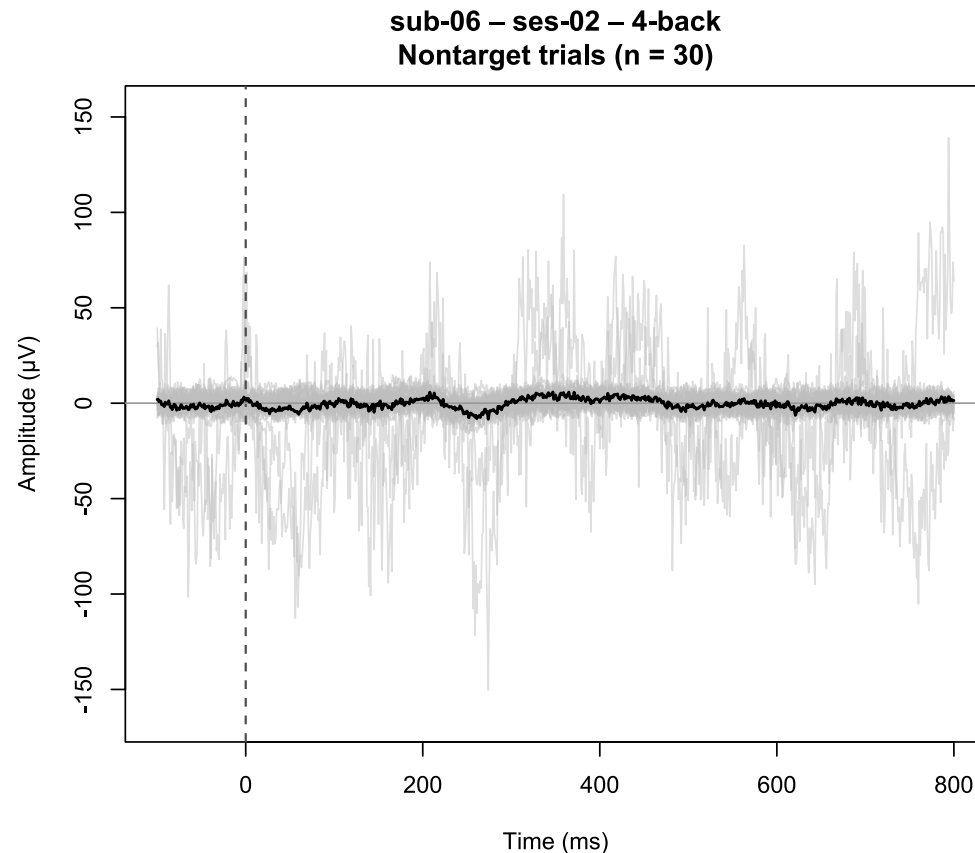
group_data.csv – ID, sex, age, training days completed

sub-01_ses-01_task-nback_behav.csv

sub-01_ses-01_task-nback_eeg.csv

To-dos:

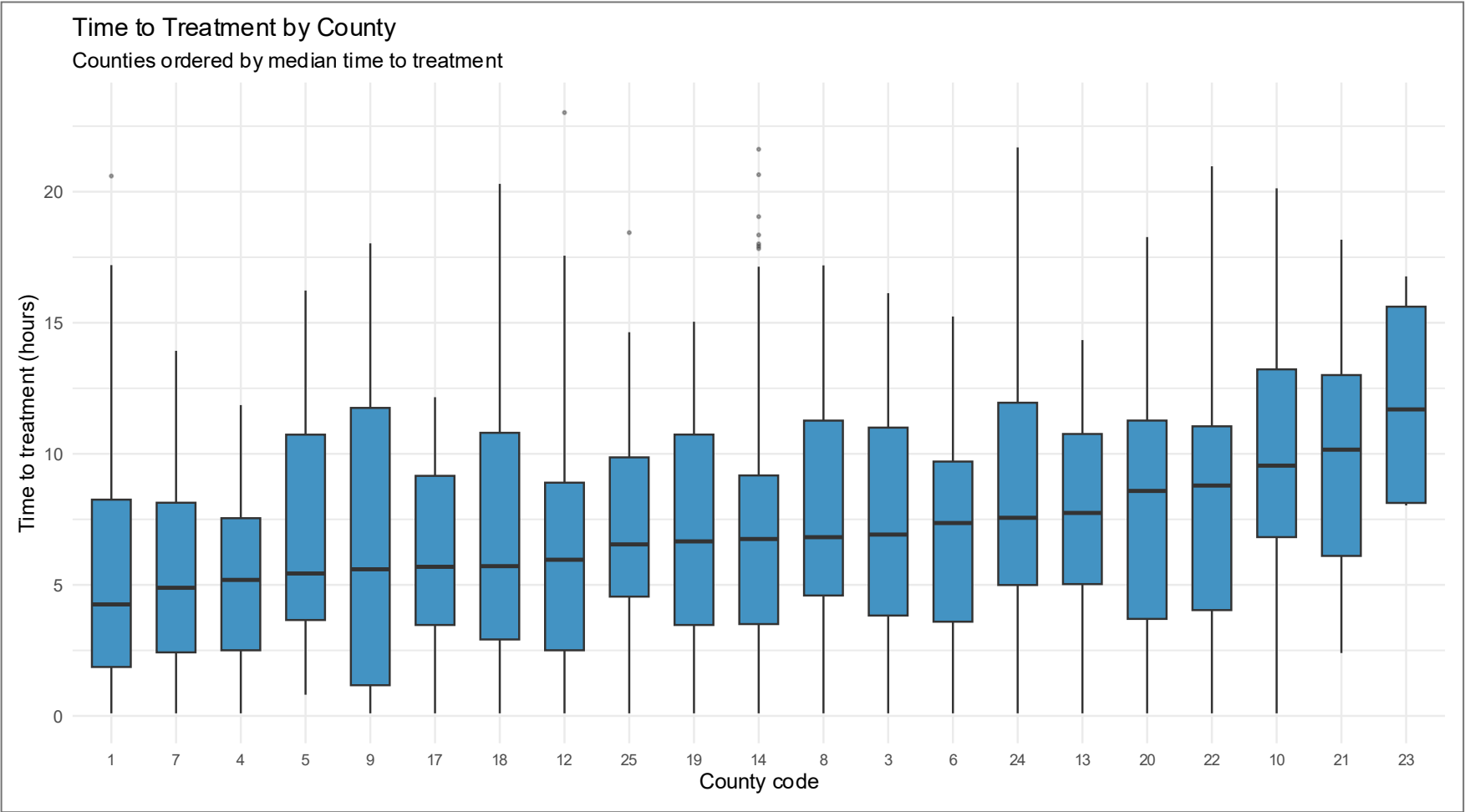
- Aggregate subject+session data
- Are all trials valid?
- Calculate amplitude response from ERP
- Relationship between EEG and training?
- Relationship between EEG and performance?

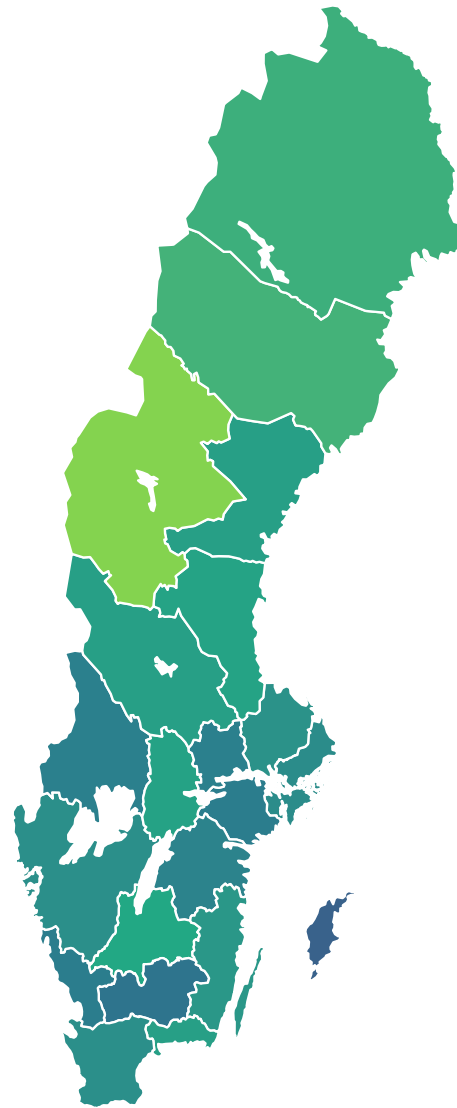


Dataset 3

Heart attack events in Sweden 2023-2025

Sex, birthdate, height, weight, bloodpressure, physical activity, smoking, alcohol, incident date, time to treat, incident outcome, death date





Dataset 4

Items:

simulated-dataset-4-sweden.csv

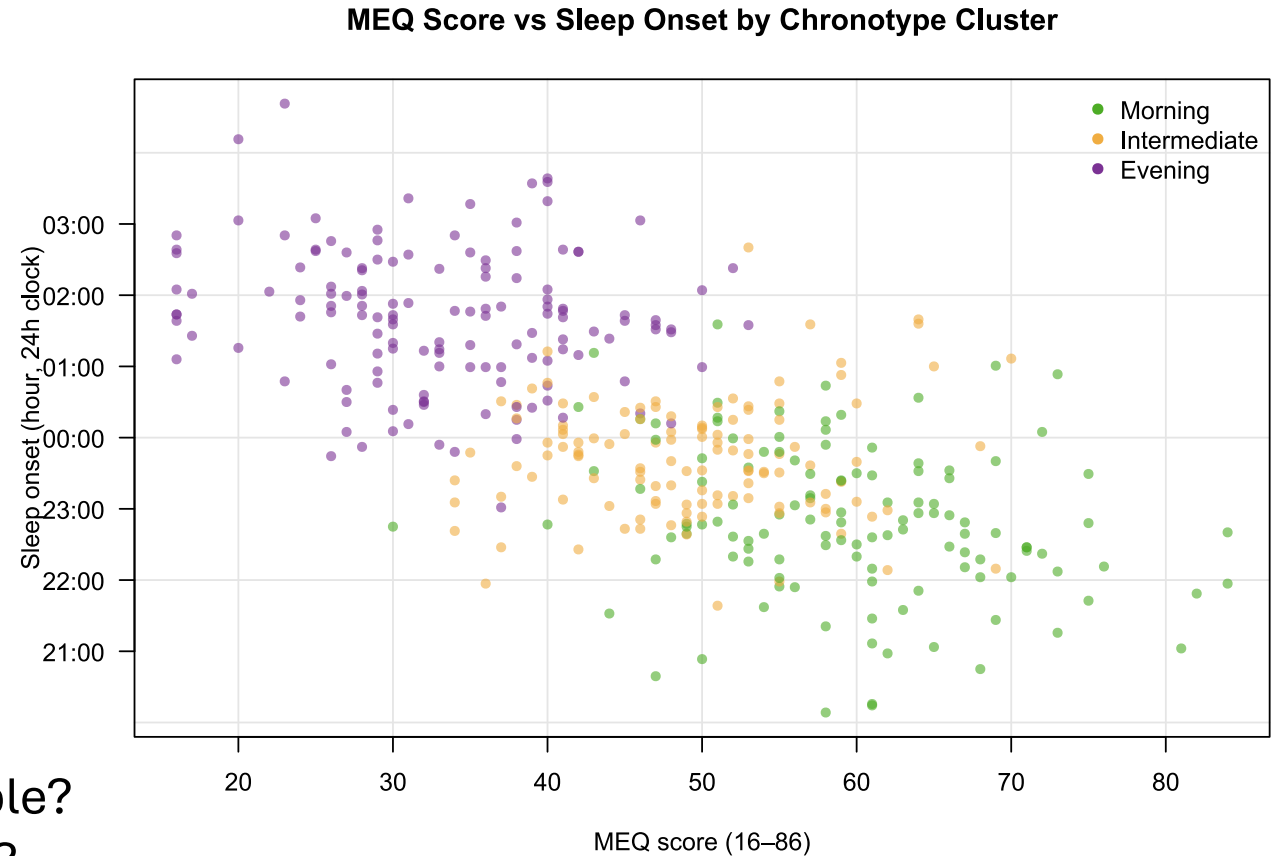
simulated-dataset-4-spain.csv

dataset-4-sweden-key.csv

dataset-4-spain-key.csv

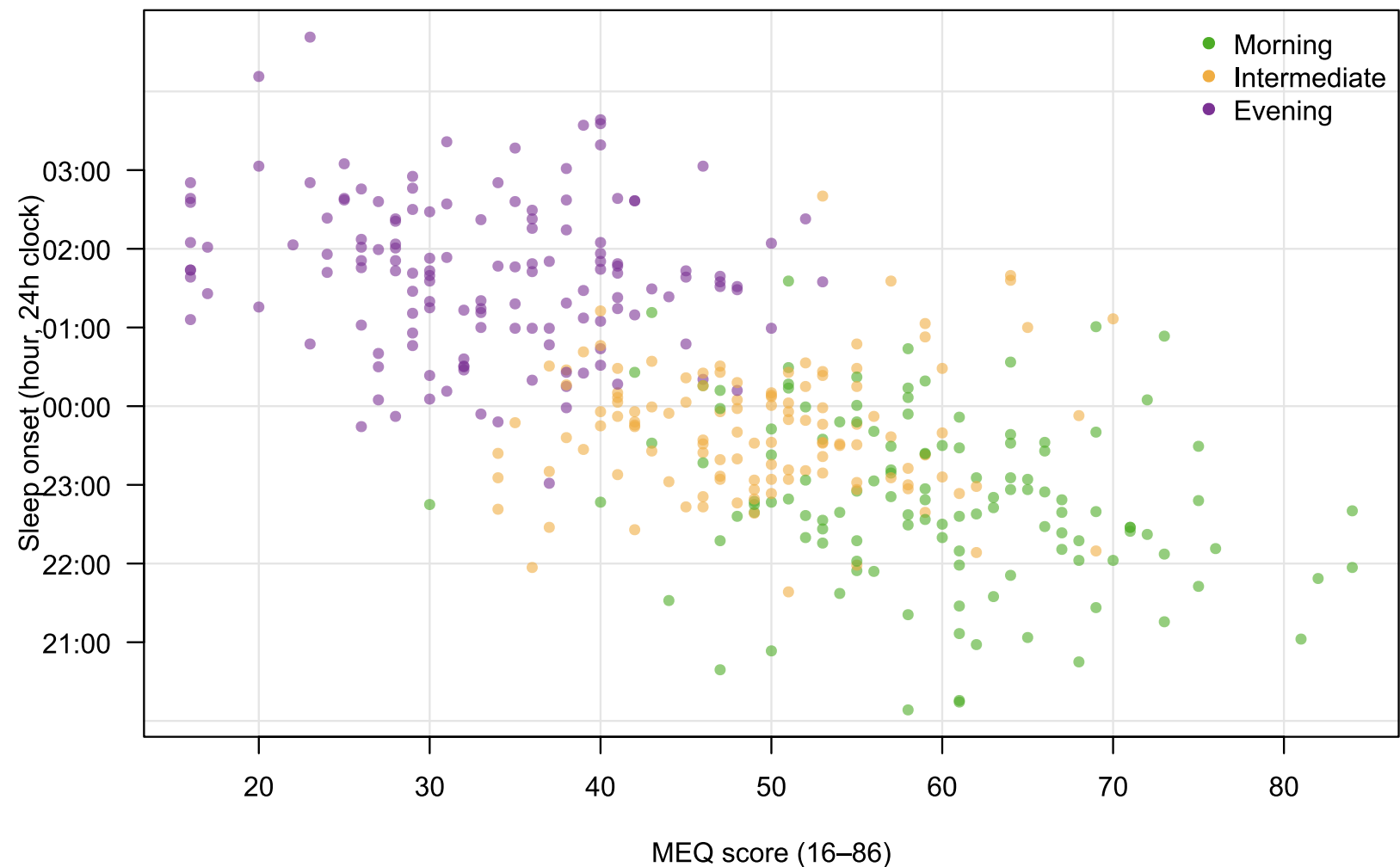
To-dos:

- Merge datasets
- Clean up
- Are Swedish people morning people?
- Do men sleep longer than women?
- Do young people stay up longer?



Dataset 4

MEQ Score vs Sleep Onset by Chronotype Cluster



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